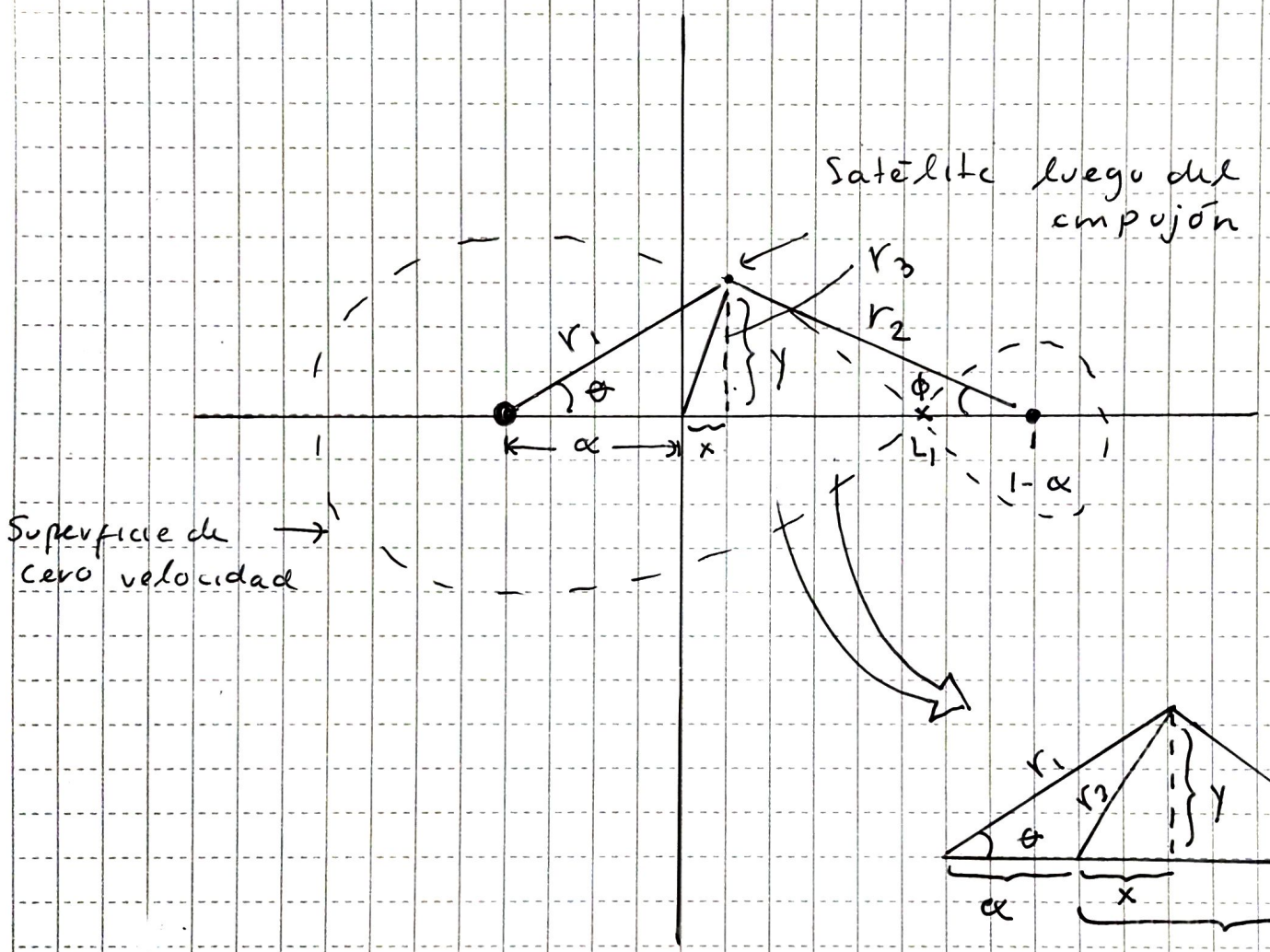

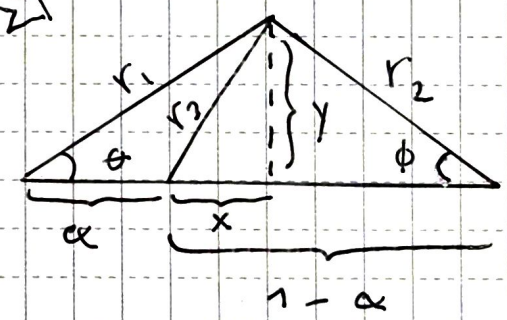

$$\cos\theta = \frac{\alpha + x}{r_1}$$

$$\cos\phi = \frac{1-\alpha-x}{r_2}$$

Aplicando ley de cosenos

$$x^2 + y^2 = r_3^2 = r_1^2 + \alpha^2 - 2\alpha(\alpha + x)$$

$$x^2 + y^2 = r_3^2 = r_2^2 + (1-\alpha)^2 - 2(1-\alpha)(1-\alpha-x)$$